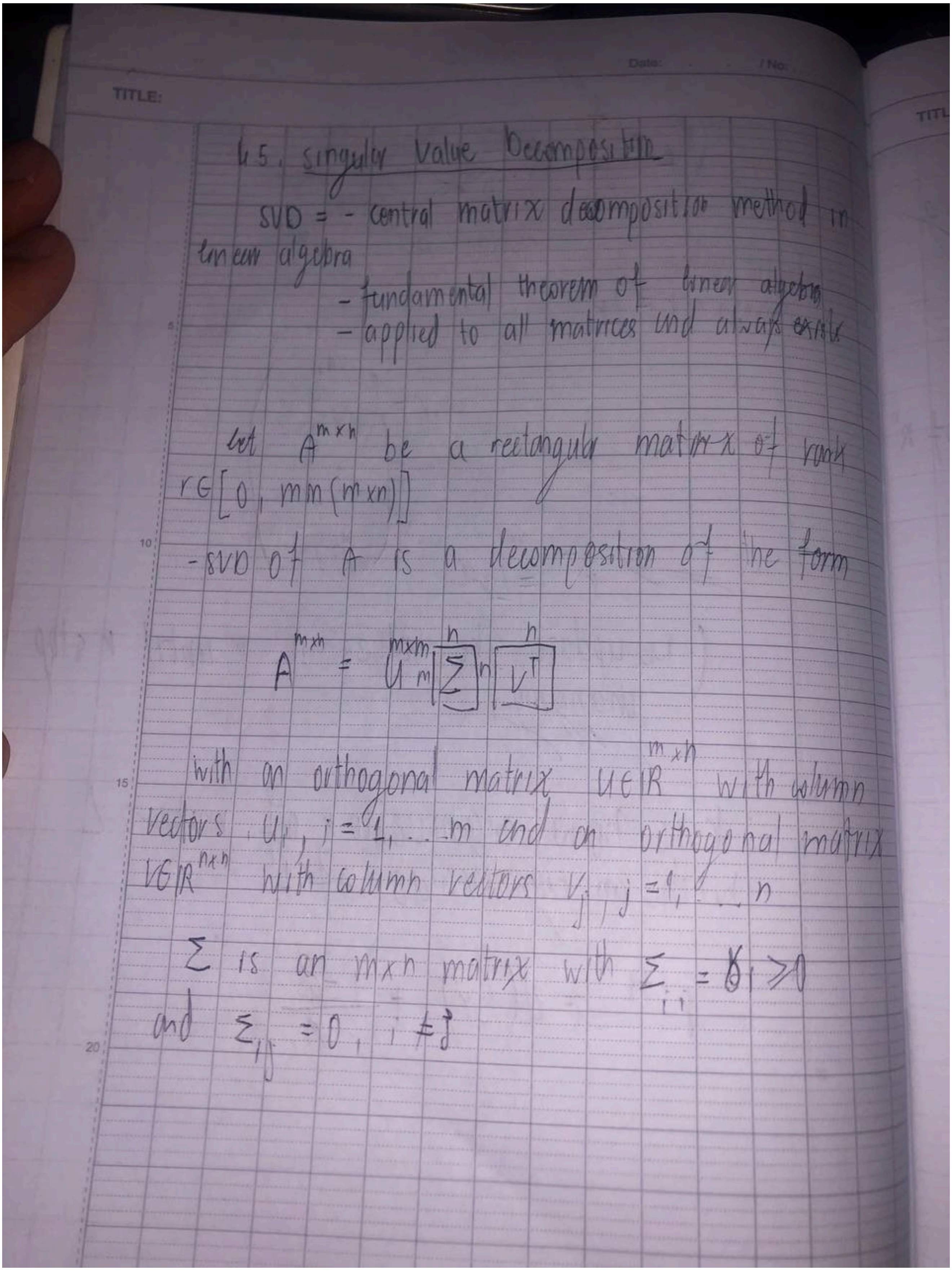
SVD = Singular Value Decomposition 1

Eigenvalue = sum of square of the distances



4.5. Singular Value Decomposition

SVD = - central matrix decomposition method in linear algebra

- fundamental theorem of linear algebra
- applied to all matrices and always exists

Let $A^{m \times n}$ be a rectangular matrix of rank $r \in [0, \min(m, n)]$

- SVD of A is a decomposition of the form

$$A^{m\times n} = U^{m\times m}\,\boxed{\Sigma}\,\boxed{V^T}$$

with an orthogonal matrix $U \in \mathbb{R}^{m \times n}$ with column vectors u_i, $i = 1, \dots m$ and an orthogonal matrix $V \in \mathbb{R}^{n \times n}$ with column vectors v_j, $j = 1, \dots n$

Σ is an $m \times n$ matrix with $\Sigma_{ii} = \sigma_i \geq 0$ and $\Sigma_{ij} = 0$, $i \neq j$

- Diagonal entries σ_i, $i = 1 \dots r$ of Σ - singular values; u_i - left singular vectors, v_i - right singular vectors

- singular vectors ordered as $\sigma_1 \geq \sigma_2 \geq \sigma_r \geq 0$

- Σ is unique and rectangular
- Σ same size as A

☆ If $m > n$, Σ has diagonal structure up to row n and then consist of 0^T row vectors from $n+1$ to m

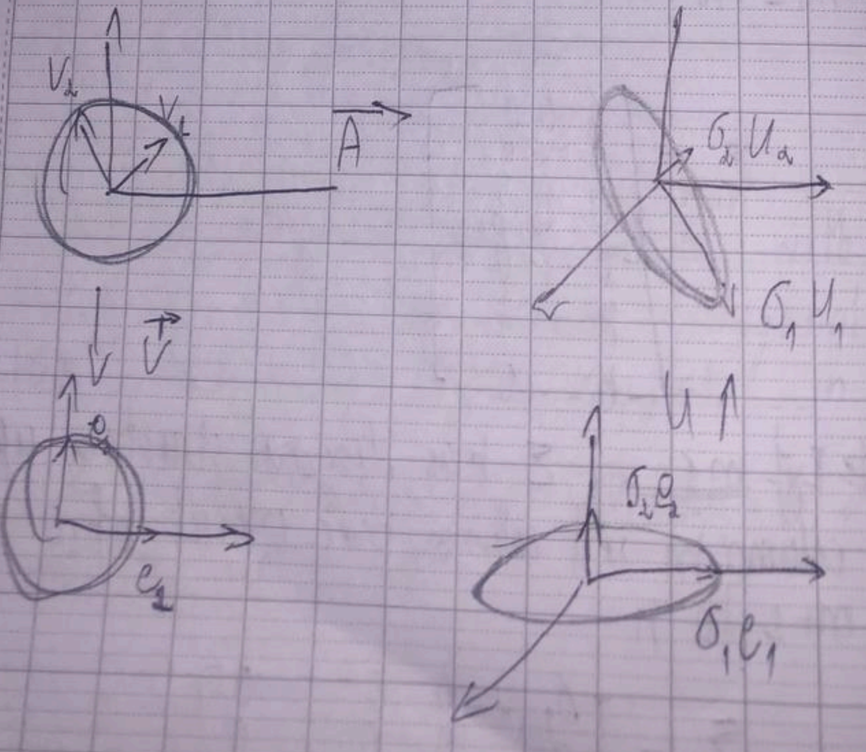
$$\Sigma = \begin{bmatrix} \sigma_1 & 0 & 0 \\ 0 & \ddots & 0 \\ 0 & 0 & \sigma_n \\ 0 & \cdots & 0 \\ \vdots & & \vdots \\ 0 & \cdots & 0 \end{bmatrix}$$

☆ If $m < n$, Σ has diagonal structure up to column m and columns that consist of 0 from $m+1$ to n

$$\Sigma = \begin{bmatrix} \sigma_1 & 0 & 0 & 0 & \cdots & 0 \\ 0 & \ddots & 0 & 0 & & 0 \\ 0 & 0 & \sigma_m & 0 & \cdots & 0 \end{bmatrix}$$

Geometric Intuition for SVD

- Interpreted as a decomposition of linear mapping $\phi: R^n \to R^m$ in to 3 operations
- Basis change via V^T followed by scaling and augumentation in dimentionality via Σ
- second basis change via U
- $\phi: R^n \to R^m$ with respect to standard bases B and C of R^n and R^m
- second basis $\hat{B}$ of R^n and $\tilde{C}$ of R^m
- $V^T = V^{-1}$ from B to $\hat{B}$



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- scale new coordinates by singular values σ_i
- Basis change in codomain $\mathbb{R}^m$ from $\tilde{C}$ to canonical basis of $\mathbb{R}^m$
- In SVD, change in basis in both domain and codomain

- Eigendecomposition - within same vector space where same basis change applied and undone
- 2 different bases simultaneously linked by Σ

Date: . . / No:

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Ex1: consider a mapping of a square grid of vector $x \in \mathbb{R}$ that fit in a box of size 2×2 centered at the origin. using the standard basis, we map these vectors using

$$A = \begin{bmatrix} 1 & -0.8 \\ 0 & 1 \\ 1 & 0 \end{bmatrix} = U \Sigma V^T$$

$$= \begin{bmatrix} -0.79 & 0 & -0.62 \\ 0.38 & -0.78 & -0.49 \\ -0.48 & -0.62 & 0.62 \end{bmatrix} \begin{bmatrix} 1.62 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} -0.78 & 0.62 \\ -0.62 & -0.78 \end{bmatrix}$$

start with x Apply $V^T \in \mathbb{R}^{2 \times 2}$

map using Σ to codomain $\mathbb{R}^3$ vector lie in $x_1 - x_2$ plane. Third coordinate always 0

Transformation of x by $U \Sigma V^T$

TITLE:

+ Construction of SVD

SVD of general matrix share some similarities with eigendecomposition of square matrix.

compare eigendecomposition of an SPD matrix

$$S = S^T = PDP^T$$

with corresponding SVD

$$S = U \Sigma V^T$$

If we set,

$$U = P = V, \quad D = \Sigma$$

SVD of SPD matrices is their eigendecomposition

computing SVD of $A \in \mathbb{R}^{m \times n}$ is equivalent to finding two sets of orthonormal bases $U = (u_1, \dots u_m)$ and $V = (v_1, \dots v_n)$ of codomain $\mathbb{R}^m$ and domain $\mathbb{R}^n$.

TITLE:

Date: / No:

Construct

start with orthonormal set of right singular vectors, $v_1, \ldots v_n \in \mathbb{R}^n$. Then left singular vectors $u_1, \ldots u_m \in \mathbb{R}^m$.

Construct symmetric, positive, semidefinite matrix $A^T A \in \mathbb{R}^{n \times m}$ from any rectangular matrix $A \in \mathbb{R}^{m \times n}$

Always diagonalize $A^T A$ and obtain

$$A^T A = PDP^T = P \begin{bmatrix} \lambda_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \lambda_n \end{bmatrix} P^T$$

$P \rightarrow$ orthogonal matrix

$\lambda_1 \geq 0$ - eigenvalues of $A^T A$

if SVD of A exist,

$$A^T A = (U \Sigma V^T)^T (U \Sigma V^T)$$

$$= V \Sigma^T U^T U \Sigma V^T$$

where U, V = orthogonal matrices

$$U^T U = I$$

$$A^T A = V \Sigma^T \Sigma V^T$$

$$= V \begin{bmatrix} \sigma_1^2 & 0 & 0 \\ 0 & \ddots & 0 \\ 0 & 0 & \sigma_n^2 \end{bmatrix} V^T$$

on comparing, $V^T = P^T$

$\sigma_i^2 = \lambda_i$

- eigen vectors of $A^TA \rightarrow$ right singular vector V of A
- eigen values of A^TA

$\rightarrow$ squared singular values of Σ

Left singular values:

SVD of symmetric matrix $AA^T \in \mathbb{R}^{m \times m}$

$$AA^T = (U\Sigma V^T)(U\Sigma V^T)^T$$

$$= U\begin{bmatrix} \sigma_1^2 & 0 & 0 \\ 0 & \ddots & \\ 0 & 0 & \sigma_m^2 \end{bmatrix} U^T$$

By spectral theorem, $AA^T = SDS^T$

Images of V_i under A have to be orthogonal

Inner product between AV_i and AV_j must be zero for $i \neq j$

$$(AV_i)^T(AV_j) = V_i^T(A^TA)V_j = V_i^T(\lambda_j V_j)$$

$$= \lambda_j V_i^T V_j = 0$$

for $m \geq r$, $AV_1, \ldots AV_r$ is the basis of r dimensional subspace of $\mathbb{R}^m$

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left singular vectors - orthonormal

Normalize images of right singular vectors AV_i and obtain

$$u_i = \frac{AV_i}{\|AV_i\|} = \frac{1}{\sqrt{\lambda_i}} AV_i = \frac{1}{\sigma_i} AV_i$$

<u>singular value equation:</u>

$$AV_i = \sigma_i u_i \ , \quad i = 1, \ldots r$$

$\forall\, i > n,\ AV_i = 0$

v_i as columns of V and u_i as column of U yield $AV = U\Sigma$

Right multiplying with V^T

$$A = U\Sigma V^T \text{ (SVD of } A\text{)}$$

TITLE: Date: / No:

Q, find the SVD of $A = \begin{bmatrix} 1 & 0 & 1 \\ -2 & 1 & 0 \end{bmatrix}$

right singular vectors v_i, singular values σ_i and left singular vectors u_i

step 1: right singular vectors as eigenbasis of A^TA

$$A^TA = \begin{bmatrix} 1 & -2 \\ 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 \\ -2 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 5 & -2 & 1 \\ -2 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

~~Compute~~ $\begin{bmatrix} 5 & -2 & 1 \\ -2 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$

$$\det(A^TA - \lambda I) = 0$$

$$\begin{vmatrix} 5-\lambda & -2 & 1 \\ -2 & 1-\lambda & 0 \\ 1 & 0 & 1-\lambda \end{vmatrix} = 0$$

$$\Rightarrow (5-\lambda)\left[1-2\lambda+\lambda^2\right] + 2\left[-2+2\lambda\right] - 1 + \lambda = 0$$

$$\Rightarrow -\lambda(\lambda-6)(\lambda-1)=0$$

- eigen values are 6, 1, 0

- <u>when $\lambda=6$</u>

$$\begin{bmatrix} -1 & -2 & 1 \\ -2 & -5 & 0 \\ 1 & 0 & -5 \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}=0$$

gaussian elimination

$$\Rightarrow \begin{bmatrix} -1 & 0 & 5 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix} \Rightarrow \begin{matrix} -x_1+5x_3 \\ x_2+2x_3=0 \end{matrix}$$

let $x_3=k$

$$\Rightarrow x_2=-2k,\quad x_1=5k$$

$$\begin{bmatrix} 5k \\ -2k \\ k \end{bmatrix}=\begin{bmatrix} 5 \\ -2 \\ 1 \end{bmatrix}k$$

Normalized eigen vector

$$\begin{bmatrix} 5 \\ 2 \\ 1 \end{bmatrix}\times\frac{1}{\sqrt{5^2+2^2+1^2}}=\frac{1}{\sqrt{30}}\begin{bmatrix} 5 \\ 2 \\ 1 \end{bmatrix}$$

TITLE:

When $\lambda = 1$

$$\begin{bmatrix} 4 & -2 & 1 \\ -2 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$$\Rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -1/2 \\ 0 & 0 & 0 \end{bmatrix} \Rightarrow \quad x_1 = 0$$

$$x_2 - \frac{1}{2} x_3 = 0$$

$$x_2 = \frac{1}{2} x_3$$

let $x_3 = k \Rightarrow x_2 = \frac{1}{2} k \Rightarrow \begin{bmatrix} 0 \\ \frac{1}{2}k \\ k \end{bmatrix} = \begin{bmatrix} 0 \\ \frac{1}{2} \\ 1 \end{bmatrix} k = \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix} k$

normalized eigen vector

$$\frac{1}{\sqrt{5}} \begin{bmatrix} 0 \\ 1 \\ 2 \end{bmatrix}$$

When $\lambda = 0$

$$\Rightarrow \frac{1}{\sqrt{6}} \begin{bmatrix} -1 \\ -2 \\ 1 \end{bmatrix}$$

now $V = P = \begin{bmatrix} 5/\sqrt{30} & 0 & -1/\sqrt{6} \\ -2/\sqrt{30} & \frac{1}{\sqrt{5}} & -2/\sqrt{6} \\ 1/\sqrt{30} & 2/\sqrt{5} & 1/\sqrt{6} \end{bmatrix} \cancel{\begin{bmatrix} 6 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}}$

TITLE:

$$\begin{bmatrix} 5/\sqrt{30} & 0 & \frac{-1}{\sqrt{6}} \\ -2/\sqrt{30} & 1/\sqrt{5} & -2/\sqrt{6} \\ 1/\sqrt{30} & 2/\sqrt{5} & 1/\sqrt{6} \end{bmatrix} \begin{bmatrix} 6 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 5/\sqrt{30} & -2/\sqrt{30} & 1/\sqrt{30} \\ 0 & 1/\sqrt{5} & 2/\sqrt{5} \\ \frac{-1}{\sqrt{6}} & -2/\sqrt{6} & 1/\sqrt{6} \end{bmatrix}$$

as singular value σ_i are square roots of eigen values of A^TA, we get them from ①
since $rk(A) = 2$, there are only two non zero singular values: $\sigma_1 = \sqrt{6}$, $\sigma_2 = 1$

The singular value matrix must be the same as A and we obtain

$$\Sigma = \begin{bmatrix} \sqrt{6} & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

step 3: left singular vectors as normalized image of right singular vectors

$$u_1 = \frac{1}{\sigma_1} A v_1 = \frac{1}{\sqrt{6}} \begin{bmatrix} 1 & 0 & 1 \\ -2 & 1 & 0 \end{bmatrix} \begin{bmatrix} 5/\sqrt{30} \\ -2/\sqrt{30} \\ 1/\sqrt{30} \end{bmatrix} = \frac{\frac{1}{\sqrt{30}} \begin{bmatrix} 6 \\ -12 \end{bmatrix}}{\sqrt{6}}$$

$$= \frac{6}{\sqrt{30}} \times \begin{bmatrix} 1 \\ -2 \end{bmatrix} \times \frac{1}{\sqrt{6}}$$

$$= \frac{1}{\sqrt{5}} \begin{bmatrix} 1 \\ -2 \end{bmatrix}$$

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$$u_2 = \frac{1}{\sigma_2} A v_2 = 1 \begin{bmatrix} 1 & 0 & 1 \\ -2 & 1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1/\sqrt{5} \\ 2/\sqrt{5} \end{bmatrix} = \frac{1}{\sqrt{5}} \begin{bmatrix} 2 \\ 1 \end{bmatrix}$$

☆ $$U = [u_1, u_2] = \frac{1}{\sqrt{5}} \begin{bmatrix} 1 & 2 \\ -2 & 1 \end{bmatrix}$$

Eigenvalue Decomposition vs SVD

eigen decomposition $\Rightarrow A = PDP^{-1}$

SVD $\quad A = U \Sigma V^T$

- SVD exist for any matrix
- eigenvalue decomposition exist only for square matrices
- vectors in eigen decomposition matrix P not necessarily orthogonal whereas U and V in SVD are orthonormal

both are composition of three linear mapping

In SVD, U and V generally not inverse of each other and in eigen decomposition P and P^{-1} are inverse of each other.

TITLE: Date: / No:

In SVD entries in diagonal matrix Σ are all real and non negative, not true in eigendecomposition

SVD and eigen decomposition are closely related through their projections

→ left singular vector of A are eigen vectors of AA^T

→ right singular vectors of A are eigen vectors of A^TA

→ Non Zero singular values of A are square roots of non zero eigen values of AA^T and equal to non zero eigen values of A^TA

Date: / No:

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